

Latin School of Chicago

COMPUTER SCIENCE



Computer Science

Have you ever wondered how your favorite apps are created, how games are designed, or how data influences the world? Computer science allows you to explore the core of technology, turning your ideas into reality. It's not just about coding; it's about solving puzzles, creating exciting projects, and understanding how technology shapes the future.

In our computer science electives, you will learn valuable skills that set you apart, from building apps and games to exploring cybersecurity and artificial intelligence. Imagine creating something that millions of people could use or finding innovative solutions to global challenges. Whether you aspire to a career in tech or simply want to explore something new, computer science offers a gateway to endless possibilities.

Take the leap—unleash your creativity and shape the future. Join us in computer science!



Computer Science Principles 1: Foundations



This course introduces students to computer science as a powerful tool for problem solving, creativity, and exploration. The course begins with understanding fundamental skills of problem solving and computational thinking and leads students towards exploring programming as a powerful tool for converting curiosity and creativity into digital artifacts. Through hands-on, project-based assignments, students can explore how computer science applies to other subjects and apply their new skills to projects that meet their interests. Students will be exposed to several programming languages and gain fluency in the foundational skills that are required in all computer science domains. This course serves as the gateway course to rest of the Computer Science sequence at Latin and is suitable for most students with no Computer Science experience or Middle School level experience.

Prerequisites: None.
Open to Grades 9, 10, 11



Computer Science Principles 2: Applications



The second half of the Computer Science Principles sequence, this course takes a deeper dive into the discipline of programming and its application as a tool for problem solving, knowledge creation, and personal expression. Building on the foundational knowledge from CSP1, students will learn how to create larger programs that utilize new techniques, libraries, data, and programming tools. Students will create interactive web applications, investigate real-world data, and research technological innovations of their own choosing as they learn more about how computational tools empower thinkers and creators across all fields and disciplines.

Prerequisites: CSP1 or department approval
Open to Grades 9, 10, 11, 12



Principles of Programming Languages



Principles of Programming Languages is a course for students who have completed CSP1 and CSP2, and are looking to learn more about the uses and design of different programming languages. Students will learn how to create solutions in both object-oriented and functional programming design techniques, with an emphasis on algorithmic approaches to solving problems. Students will use a wide variety of programming languages to explore other advanced topics in programming, such as data structures and programming language design. Students should expect to apply their previous understandings of core computational concepts to reading and writing code in a variety of languages. Through this study, the introduction of new concepts, and open ended projects, students will improve their technical programming skills. Through a series of projects, students will gain experience in designing, programming, testing, debugging, and documenting larger scale programs.

Prerequisites: CSP1 & CSP2 or department approval
Open to Grades 9, 10, 11, 12



Modern Computers From NAND to Tetris



In this course, students will take a deep dive to understand how computers work, from the smallest bits of electronic hardware in their computers all the way up to the applications they use every day. Students will learn about the design and implementation of both hardware and software through studying computer design and then writing and simulating each step on their own. The course will help you come to understand the various layers of abstraction that come together to make modern computers work through a series of guided projects. This course requires some programming, but has a big focus on problem solving, design, and critical thinking skills without code. This course is inspired by a course by Noam Nisan and Shimon Schocken, for which additional details can be found online at <http://nand2tetris.org>.

Prerequisites: CSP2 or department approval

Open to Grades 10, 11, 12

*Offered in alternating years. Offered upcoming school year – 2023-2024.



Algorithms and Data Structures



This course aims to help students understand and reason about the programming topics that they were introduced to in previous courses. Students will learn about the fundamental data structures in computer science and the algorithms that work with them, including structures like stacks, queues, tables, trees, and graphs. Along the way, students will learn how to analyze the space and time complexity of algorithms, giving them a powerful tool for reasoning about which data structures are more efficient than others for specific applications. Each algorithm covered will be motivated by its applications to mathematics, science, or engineering disciplines. This course requires a strong foundation in programming and logical thinking skills and will prepare students for college or career level programming tasks.

Prerequisites: Principles of Programming Languages or department approval

Department: Computer Science
Open to Grades 10, 11, 12

